

Notes on Holographic Entanglement Entropy

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Abstract

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1 Introduction

2 RT Formula

2.1 Ryu-Takayanagi Formula

Theorem 2.1. $S(A) = \frac{\text{area}(\gamma)}{4G}$ and γ satisfies the following condition:

1. $\partial\gamma = \partial A$.
2. γ and A are in the same time slice.
3. the area of γ take extreme value among all the submanifold satisfying condition 1 and 2.

Theorem 2.2. $S(A) = \frac{\text{area}(\gamma)}{4G}$ and γ satisfies the following condition:

1. $\partial\gamma = \partial A$
2. γ is homologous to A (i.e. there exists a n -dimensional spacelike submanifold in AdS_{n+1} whose boundary is $\gamma \cup A$)
3. the area of γ take extreme value among all the submanifold satisfying condition 1 and 2

Theorem 2.3. In AdS_3/CFT_2 , the HRT surface is a spacelike geodesic in AdS space.

2.2 Hubeny-Rangamani-Takayanagi Formula

2.3 Proofs of RT Formula and HRT Formula

2.4 Quantum Extremal Surface

2.5 Camps-Dong Formula

2.6 Membrane Theory

2.7 Bit Threads

2.8 Wilson Line

2.9 Modular Flow

2.10 Entanglement Contour

2.11 Entanglement Wedge Cross Section

3 HEE in Time-Independent CFT

3.1 Interval in 1+1-dimensional CFT

3.1.1 Interval in CFT on $\mathbb{R}^{1,1}$

根据 AdS/CFT 对偶，CFT 的中心荷 c 与 AdS 空间的 AdS 半径 L 的关系为 [?]

$$c = \frac{3L}{2G_N^3} \quad (1)$$

采用 Poincare 坐标，边界是 $z = 0$ 的超曲面。考虑区间 $A = [-\frac{l}{2}, \frac{l}{2}]$ 。由对称性可知端点为 $x = -\frac{l}{2}$ 和 $x = \frac{l}{2}$ 的测地线关于中点 $x = 0$ 对称。方便起见取 $L = 1$ 。设测地线为 $z(x)$ ， z 的最大值为 z^* ，则测地线的长度为：

$$2 \int_0^{z^*} dz \frac{\sqrt{1 + (z')^2}}{z} \quad (2)$$

通过变分得到测地线方程为 $x(z) = -\sqrt{l^2/4 - z^2}$ 。因为上面的积分在 $z \rightarrow 0$ 时发散，需引入 UV 截断 $z \geq a$ 。因此测地线长度为：

$$l = 2 \int_a^l dz \frac{l}{z\sqrt{l^2 + 4z^2}} = 2 \ln \left(\frac{l}{a} \right) \quad (3)$$

带入 RT 公式，得到纠缠熵为

$$S_A = \frac{c}{3} \ln \left(\frac{l}{a} \right) \quad (4)$$

考虑 CFT 经过足够长时间演化后达到热平衡。和有限温 CFT 对偶的是 BTZ 黑洞，其度规为：

$$ds^2 = -(r^2 - M) dt^2 + \frac{dr^2}{r^2 - M} + r^2 dx^2 \quad (5)$$

其中 m 为黑洞质量，视界位于 $r = r_H = \sqrt{M}$ ，对应的霍金温度为 $\beta^{-1} = \frac{\sqrt{m}}{2\pi}$ 。改用 Poincare 坐标，度规为：

$$ds^2 = \frac{1}{z^2} (-(1 - Mz^2) dt^2 + \frac{dz^2}{1 - Mz^2} + dx^2) \quad (6)$$

选择测地线的长度 λ 作为测地线的参数。通过变分可以得到测地线满足的方程，

$$-r_H E = -(r^2 - r_H^2) \dot{t} \quad (7)$$

$$r_H J = r^2 \dot{x} \quad (8)$$

$$1 = -(r^2 - r_H^2) \dot{t}^2 + \frac{\dot{r}^2}{r^2 - r_H^2} + r^2 \dot{x}^2 \quad (9)$$

其中 E 和 J 是待定系数。可以解出测地线方程为 [?]

$$r(\lambda) = \frac{r_H}{2} \sqrt{[e^\lambda + ((J+1)^2 - E^2)e^{-\lambda}][e^\lambda + ((J-1)^2 - E^2)e^{-\lambda}]} \quad (10)$$

$$t(\lambda) = t_0 + \frac{1}{2r_H} \ln \left| \frac{e^\lambda + (J^2 - (1+E)^2)e^{-\lambda}}{e^\lambda + (J^2 - (1-E)^2)e^{-\lambda}} \right| \quad (11)$$

$$x(\lambda) = x_0 + \frac{1}{2r_H} \ln \left(\frac{e^\lambda + ((J-1)^2 - E^2)e^{-\lambda}}{e^\lambda + ((J+1)^2 - E^2)e^{-\lambda}} \right) \quad (12)$$

带入边界条件可以得到

$$E = 0, \quad J = \cosh\left(\frac{r_H l}{2}\right) \quad (13)$$

可以求出测地线长度为

$$l = 2 \ln \left(\frac{\beta}{\pi a} \sinh \left(\frac{\pi l}{\beta} \right) \right) \quad (14)$$

带入 RT 公式，得到纠缠熵为

$$S_A = \frac{c}{3} \ln \left(\frac{\beta}{\pi a} \sinh \left(\frac{\pi l}{\beta} \right) \right) \quad (15)$$

这两个静态结果在后文中将分别对应热化过程中含时演化的纠缠熵的初始极限和热化后极限，在计算含时演化情况下的纠缠熵时也需要参考这两个结果。

3.1.2 Interval in CFT on $\mathbb{R} \times S^1$

4 Quantum Quench

5 HEE in Time-Dependent CFT

5.1 Vaidya/AdS Spacetime

AdS spacetime is the solution of Einstein's equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (16)$$

when $T_{\mu\nu} = 0$ and .

Let the dimension of spacetime be d . For convenience, Let $8\pi G = 1$, $c = 1$ and $\Lambda = -\frac{d(d+1)}{2L^2}$.

Metric of pure AdS Spacetime in global coordinates (t, ρ, θ) is

$$ds^2 = L^2(-\cosh^2(\rho)dt^2 + d\rho^2 + \sinh^2(\rho)d\theta^2) \quad (17)$$

其中, $t \in \mathbb{R}$, $\rho \in (0, \infty)$ and $\theta \in [0, 2\pi)$.

Let $r = L \sinh(\rho)$. Metric of pure AdS Spacetime in static coordinates (t, r, θ) is

$$ds^2 = -(1 + \frac{r^2}{L^2})dt^2 + (1 + \frac{r^2}{L^2})^{-1}dr^2 + r^2d\theta^2 \quad (18)$$

其中, $t \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of pure AdS Spacetime in Eddington-Finkelstein coordinates (v, r, θ) is

$$ds^2 = -(1 + \frac{r^2}{L^2})dv^2 + 2dvdr + r^2d\theta^2 \quad (19)$$

其中, $v \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of pure AdS Spacetime in Poincare coordinates is

$$ds^2 = \frac{L^2}{z^2}(-dt^2 + dx^2 + dz^2) \quad (20)$$

其中, $t \in \mathbb{R}$, $x \in \mathbb{R}$ and $z \in (0, \infty)$.

Metric of pure AdS Spacetime in Poincare-Eddington-Finkelstein coordinates (v, z, x) is

$$ds^2 = \frac{L^2}{z^2}(-dv^2 + 2dvdx + dx^2) \quad (21)$$

其中, $v \in \mathbb{R}$, $x \in \mathbb{R}$ and $z \in (0, \infty)$.

the static, Shere balck hole in AdS spacetime is described by BTZ black hole.

Metric of BTZ black hole in static coordinates (t, r, θ) is

$$ds^2 = -(-M + \frac{r^2}{L^2})dt^2 + (-M + \frac{r^2}{L^2})^{-1}dr^2 + r^2d\theta^2 \quad (22)$$

其中, $t \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of BTZ black hole in Eddington-Finkelstein coordinates (v, r, θ) is

$$ds^2 = -(-M + \frac{r^2}{L^2})dv^2 + 2dvdr + r^2d\theta^2 \quad (23)$$

其中, $v \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of BTZ black hole in Poincare-Eddington-Finkelstein coordinates (v, z, x) is

$$ds^2 = \frac{L^2}{z^2}(-(1 - Mz^2)dv^2 + 2dvdx + dx^2) \quad (24)$$

其中, $v \in \mathbb{R}$, $x \in \mathbb{R}$ and $z \in (0, \infty)$.

Vaidya Metric is used to describe the process of the collapse of a thin shell to a BTZ black hole.

Metric of Vaidya/AdS spacetime in static coordinates (t, r, θ) is

$$ds^2 = -(-m(v) + \frac{r^2}{L^2})dt^2 + (-m(v) + \frac{r^2}{L^2})^{-1}dr^2 + r^2d\theta^2 \quad (25)$$

其中, $t \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of Vaidya/AdS spacetime in Eddington-Finkelstein coordinates (v, r, θ) is

$$ds^2 = -(-m(v) + \frac{r^2}{L^2})dv^2 + 2dvdr + r^2d\theta^2 \quad (26)$$

其中, $v \in \mathbb{R}$, $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$.

Metric of Vaidya/AdS spacetime in Poincare-Eddington-Finkelstein coordinates (v, z, x) is

$$ds^2 = \frac{L^2}{z^2}(-(1 - m(v)z^2)dv^2 + 2dvdx + dx^2) \quad (27)$$

其中, $v \in \mathbb{R}$, $x \in \mathbb{R}$ and $z \in (0, \infty)$.

5.2 Time-Dependent CFT after Global Quench

下面计算单区间纠缠熵在 Vaidya-BTZ 背景中的结果。多区间纠缠熵和更高阶信息度量最终都要通过 $AdS/Vaidya$ 背景中单条测地线长度函数 $L(l, t)$ 计算得到。考虑由 Vaidya-BTZ 度规描述的 2+1 维 AdS 时空:

$$ds^2 = \frac{L^2}{z^2}(-(1 - M\theta(v)z^2)dv^2 + 2dvdx + dx^2) \quad (28)$$

其中 $\theta(v)$ 是阶跃函数。

当 $t \leq 0$ 时，度规为纯 AdS 度规。此时测地线与纯 AdS 情况下的结果相同。为方便后面的数值计算，我们把测地线长度中 UV 截断的部分 $2\ln(2a)$ 去掉，把测地线的长度写为

$$L(l, t) = 2 \ln \frac{l}{2} \quad (29)$$

当 t 足够大时，度规为 BTZ 度规。此时测地线与 BTZ 黑洞情况下的结果相同。

$$L(l, t) = 2 \ln \frac{\sinh\left(\frac{r_H l}{2}\right)}{r_H} \quad (30)$$

在一般的 t 时刻，测地线会穿过薄球壳。在球壳外，测地线方程的解和 BTZ 黑洞的情况相同，同样将测地线中的参数设为 E 和 J 。在球壳内，测地线方程的解和纯 AdS 时空的情况相同。设 t 时刻球壳的半径是 $r_0(t)$ 。 $r_0(t)$ 由下面的方程决定

$$0 = v(r_0) = \frac{1}{2r_H} \log \left[\frac{r_0 - r_H r_0^2 - (1 + E)r_H^2 \pm \sqrt{r_0^4 - (1 + J^2 - E^2)r_H^2 r_0^2 + J^2 r_H^4}}{r_0 + r_H r_0^2 - (1 - E)r_H^2 \pm \sqrt{r_0^4 - (1 + J^2 - E^2)r_H^2 r_0^2 + J^2 r_H^4}} \right] + t \quad (31)$$

带入边界条件可以得到

$$E = -\frac{r_H \sqrt{r_0^2 - (\frac{2}{l})^2}}{2r_0^2} \quad \text{and} \quad J = \frac{2}{r_H l} \quad (32)$$

所以可以得到 t 时刻的测地线长度 [?]

$$L(l, t) = 2 \ln \frac{(r_0 + \sqrt{r_0^2 - (\frac{2}{l})^2})l}{2} - \ln \left[\frac{4r_0^4(2r_0^2 - (\frac{2}{l})^2 - r_H^2) + r_H^4(r_0^2 - (\frac{2}{l})^2) + 4r_0^3(2r_0^2 - r_H^2)\sqrt{r_0^2 - (\frac{2}{l})^2}}{4r_0^4} \right] \quad (33)$$

令 $c = \sqrt{1 - s^2}$ ，以及

$$2\rho = \coth(r_H t) + \sqrt{\coth^2(r_H t) - \frac{2c}{c+1}}. \quad (34)$$

$$l = \frac{1}{r_H} \left[\frac{2c}{s\rho} + \ln \left(\frac{2(1+c)\rho^2 + 2s\rho - c}{2(1+c)\rho^2 - 2s\rho - c} \right) \right], \quad (35)$$

可以将测地线长度写成如下形式，

$$L(l, t) = 2 \ln \frac{\sinh(r_H t)}{r_H s(l, t)} \quad (36)$$

所以，Vaidya/AdS 情况下的测地线长度为

$$L(l, t) = \begin{cases} 2 \ln \frac{l}{2}, & t \leq 0 \\ 2 \ln \frac{\sinh(r_H t)}{r_H s(l, t)}, & 0 < t < \frac{l}{2} \\ 2 \ln \frac{\sinh \frac{r_H l}{2}}{r_H}, & t \geq l/2 \end{cases} \quad (37)$$

纠缠熵在热化过程中随时间的演化行为见图1

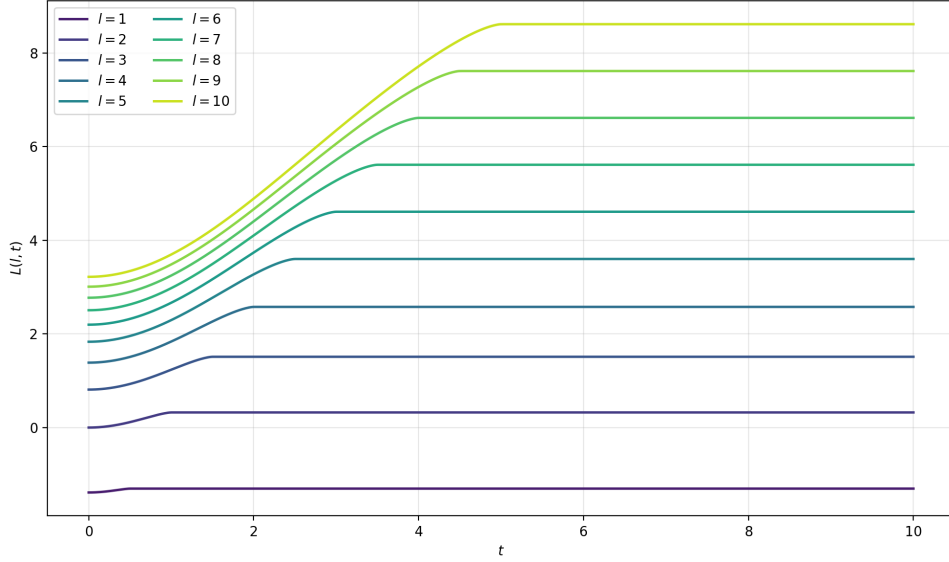


图 1: $l = 1, 2, \dots, 10$ 时 $L(l, t)$ 随时间的演化

可以从图1看出，在 l 比较大时，纠缠熵在热化过程中随时间近似线性增长 [?, ?]。

6 Multi-Partite Entanglement Entropy

6.1 HEE of Multi-Interval in Time-Independent CFT

7 N-Partite Information